

Class Activity: Implementing and Querying a Relational Database

Seth Garciano, Tristin Bates, Troy Franks and Yisong Wang

Tristin's Queries:

1. One query that demonstrates aggregation (e.g., SUM, COUNT, AVG).
 - This query counts the total appointments by using the COUNT function on every row in the appointment table
2. For each doctor, who are their two highest-billed patients?
 - This query adds the total amount billed for every patient for each doctor and then ranks the patients from highest to lowest and then only shows the top two highest-billed patients for each doctor.

Yisong's Queries:

1. Calculate the total dollar amount billed to all patients.
 - This query calculates the total amount billed to all patients, including paid, partially paid, and unpaid bills.
2. One query that uses a stored procedure or view.
 - This query uses the medical_history view to display patients' medical histories, including doctor IDs, diagnoses, treatments, and visit dates.
3. How did visits and billing change from month to month?
 - This query compares monthly visits and billing totals, showing the numerical and percentage changes from the previous month.

Seth's Queries:

1. Count how many patients visited the clinic last month.
 - This query counts the number of unique patients who visited the clinic last month by using COUNT(DISTINCT) on each patient's ID.
2. One query that uses GROUP BY with HAVING.
 - This query finds patients who visited the clinic at least twice by counting their visits and using HAVING to filter the results.
3. Which doctors bill more than the average doctor, and what share of all clinic billing is each one's?
 - This query finds doctors who billed more than the average doctor by calculating their total bills and comparing them to the average.

Troy's Queries

1. Count total amount of patients in the database
 - This query counts the total amount of patients in the patient table
2. One query that uses subqueries and JOINS (e.g., INNER, LEFT, RIGHT)
 - This query lists each patient that has completed at least one appointment with their total appointment count and total billed amount and total amount paid

Team Contribution

Team Member	Contribution
Seth Garciano	Completed the patients visited in the last month, completed GROUP BY with HAVING, completed doctors' bills average comparisons and what share of all clinic billing is each one's.
Tristin Bates	Aggregation and top 2 highest-billed patients queries, and list/explanation for each query
Troy Franks	EERD diagram and count total number of patients and one query that uses subqueries and JOINS
Yisong Wang	Build View, Database structure & data SQL and answer Q3, Q7, Q9